Unmanned Aerial Vehicle Flight Data Anomaly Detection and Recovery Prediction Based on Spatio-Temporal Correlation

Jie Zhong, Yujie Zhang, *Member, IEEE*, Jianyu Wang, Chong Luo, and Qiang Miao, *Senior Member, IEEE*

***Abstract*—With the development of unmanned aerial vehicle (UAV) technology, a UAV is gradually applied to a variety of civil fields, such as photography, power line inspection, and environmental monitoring. At the same time, the safety and reliability of a UAV also attract wide attention. Anomaly detection is one of the key technologies to improve the safety of an UAV. The structure of the UAV system is complex, and there are complex spatio-temporal correlations among the high-dimensional flight data with many parameters. However, the existing methods often ignore the spatio-temporal correlation of data and lack parameter selection, which is used to abandon the parameters without a positive impact on anomaly detection results. This article proposes a spatio-temporal correlation based long short-term memory (LSTM) method for anomaly detection and recovery prediction of UAV flight data. First, an artificial neural network correlation analysis is proposed to preliminarily mine the spatio-temporal correlation in flight data and to obtain the correlation parameter sets. Second, the LSTM model is established, and the mapping among different parameters is realized. Finally, anomaly detection and recovery prediction are carried out based on parameter sets mapping model. The effectiveness of the proposed method is verified by generating sample sets with anomaly injection on real UAV flight data.**

***Index Terms*—Anomaly detection, correlation analysis, recovery prediction, spatio-temporal correlation based long short-term memory (STC-LSTM), unmanned aerial vehicle (UAV).**

I. Introduction

With the rapid development of unmanned aerial vehicle (UAV) technology, manufacturing and operation costs of a UAV are greatly reduced, which promotes the use of UAVs in different civil fields, including UAV photography, power line inspection, environmental monitoring, and express delivery [1]–[3]. Meanwhile, more and more attention has been paid to the safety and reliability of a UAV [4]. A UAV is a complex system composed of mechanical, hydraulic, and electrical subsystems with complex software and control modules. UAV flight data are collected by sensors, which directly reflect UAV flight states. To reduce property loss and environmental damage caused by UAV accidents, it is very important to carry out anomaly detection and recovery prediction research on UAV flight data, to effectively improve the safety and reliability of UAVs.

Anomaly detection is one of the key technologies to ensure the safety of the system, and it refers to the problem of finding patterns in data that do not conform to the expected behavior [5]. Anomaly detection has a wide range of applications, such as finance, medical treatment, industry, and the Internet [6]. Recently, data-driven methods are widely used in these application scenarios with sufficient available data. According to the scale of algorithms in data-driven methods, it can be divided into shallow structure and deep structure [7]. Nonparametric learning methods, such as support vector machine (SVM) [8]–[10], one-class SVM [11], and extended isolation forest [12], can be regarded as shallow structure methods. The ability of shallow structure methods to learn high-dimensional functions is very limited. The shallow structure methods are suitable for processing low-dimensional, small sample data. The deep structure methods are neural networks with multiple hidden layers, generally composed of multiple parameterized nonlinear modules, such as convolutional neural networks [13], [14], variational autoencoder [15], and generative adversarial network [16]. The deep structure methods can perform high-level and abstract representation of data and have strong generalization capabilities.

Generally, there are three main objectives for anomaly detection of UAV flight data, which are highlighted as follows: first, detect anomaly in time to avoid serious consequences; second, locate anomaly source to guide subsequent operation and maintenance; and third, predict recovery value of anomaly point to minimize damage caused by the anomaly. In general, current UAV anomaly detection methods can be divided into three categories: knowledge-based methods, model-based methods, and data-driven methods [17], [18]. The knowledge-based methods usually require experts in the research field to summarize the anomaly detection model, which includes two processes: knowledge acquisition and knowledge engineering. Knowledge acquisition refers to obtaining solutions to

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Jie Zhong and Jianyu Wang are with the School of Aeronautics and Astronautics, Sichuan University, Chengdu 610065, China (e-mail: zhongjie_scu@outlook.com; jian.yu_wang@hotmail.com).

Yujie Zhang, Chong Luo, and Qiang Miao are with the College of Electrical Engineering, Sichuan University, Chengdu 610065, China (e-mail: zhangyj@scu.edu.cn; luochong33@foxmail.com; mqiang@scu.edu.cn).

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problems from experts. Knowledge engineering represents a large amount of knowledge into a set of precise rules that are used to construct an expert knowledge system. The main difficulty of knowledge-based methods is that expert knowledge is difficult to obtain and express. In this aspect, a knowledge-based method often requires time-consuming collaborative development between developers and experts [19]. Model-based methods usually require the establishment of an accurate physical model. The residual between the estimated value and the actual measurement is evaluated to determine whether an anomaly has occurred [20]. However, a UAV is a complex system that integrates mechanical, electronic, hydraulic, and other parts. It is very difficult to establish a complete and accurate physical model of UAVs. Data-driven methods received a lot of attention recently, which benefits from the rapid improvement of computing power and intelligent algorithms. Data-driven methods consider that data can describe the running process and state of the system. Compared with knowledge-based methods and model-based methods, data-driven methods do not require a large amount of expert knowledge or a complicated UAV system model [18]. Data-driven methods can be carried out directly by using flight data.

A UAV is a complex system with lots of sensors, so the flight data of UAV have the characteristics of high dimensionality. Due to the influence of the UAV structure and flight process, there is spatio-temporal correlation [21] within the flight data. The existing data-driven UAV anomaly detection methods can be mainly divided into two categories: temporal correlation methods and spatio-temporal correlation methods. Temporal correlation methods consider that the flight data are correlated in the time dimension, and learn the characteristics or representations of the data only from the historical data of the detected parameter. When the new data do not conform to the historical characteristics or representations, it will be judged as anomaly [2], [18], [22]. However, temporal correlation methods only consider the correlation of the data in the time dimension, and ignore the mutual influence of the data in the spatial dimension. Spatio-temporal correlation methods comprehensively consider the relationship among the data in the time dimension and the spatial dimension, use multiparameter historical data to build model, and fully explore the relationship among the data in the time dimension and the space dimension. When new data do not satisfy the relationship, it is considered as anomaly [23]–[25]. The deep structure methods are a common way for mining spatio-temporal correlation in data, and long short-term memory (LSTM) occupies an important position in processing time-series data. Benkuan *et al.* [23] used an LSTM model to mine the spatio-temporal correlation in flight data, but it relies on prior knowledge in the selection of model's input parameters. In the absence of prior knowledge, a simple solution is to directly select all parameters. However, there are many parameters in UAV flight data, and not all parameters in the flight data have a positive effect on results. Too many useless parameters will lead to many problems, such as underfitting, the curse of dimensionality, and excessive volume. In this article, the correlation methods are used to mine the spatial relationship of the flight data, and the correlation analysis results are used as the basis for selecting the input parameters of the anomaly detection model.

Correlation analysis refers to measuring the closeness of the correlation among the two variables. Commonly used correlation analysis methods include Pearson correlation coefficient (PCC) [26], time lag correlation (TLC), dynamic time warping (DTW) [27], and maximal information coefficient (MIC) [28]. PCC is the ratio between the covariance of two variables and the product of their standard deviations, which can only reflect a linear correlation of variables. TLC is a measure of the PCC of the misalignment of two time series, which can define the directionality among the two variables, such as the follow relationship. DTW is a method of calculating the path among two variables. It can minimize the distance between the two variables and process data of different lengths. MIC measures the correlations among two variables through mutual information and has nonlinear analysis capability. A UAV is a complex system, and there are also complex spatio-temporal correlations in flight data. However, the existing correlation analysis methods are weak in analyzing these correlations.

To solve the difficulty of choosing the parameters, and to achieve high-precision detection and recovery value prediction, this article proposes a spatio-temporal correlation based LSTM (STC-LSTM) deep structure data-driven method. First, artificial neural network correlation analysis (ANNCA) is used to preliminarily reflect the spatio-temporal correlation within flight data and obtain the parameter sets with correlation. Second, the STC-LSTM model is established for the parameter sets with spatio-temporal correlation, and the mapping model among different parameters is realized. Finally, anomaly detection and recovery prediction are carried out based on the parameter set mapping model. The contributions of this article are as follows.

1) A correlation analysis method ANNCA based on a fully connected neural network is proposed, which solves the difficult problem of reflect complex and nonlinear correlation in flight data.
2) An STC-LSTM anomaly detection framework for UAV flight data is proposed, which includes correlation analysis, anomaly detection, and recovery prediction. By using correlation analysis as the basis of parameter selection, the dependence on expert knowledge and the number of model parameters are reduced effectively. STC-LSTM can greatly improve the accuracy of anomaly detection and provide high-precision predictive recovery values.
3) Taking the real UAV flight data with high-dimensional and complex correlation characteristics as the object, the deep structure method is used to carry out anomaly detection research. Meanwhile, three anomalous implantation methods are used to construct anomalous samples to verify the effectiveness of the STC-LSTM.

The rest of this article is organized as follows. Section II introduces the fundamental theory and STC-LSTM framework. Section III introduces the anomaly detection and recovery prediction process and experiment in detail. Section IV summarizes the work of this article and points out the future direction.

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II. Fundamental Theory and Modeling

A. Correlation Analysis

1) Linear Correlation Analysis Based on PCC: PCC is a commonly used method for correlation analysis, which has the characteristics of simple principle and small calculation loss. PCC is suitable for finding linear correlation among flight data, which can be formulated as follows:

$$\rho(X,Y) = \frac{\mathrm{Cov}(X,Y)}{\sigma_X \sigma_Y}. \tag{1}$$

where σ_X and σ_Y are the standard deviations of variables X and Y , respectively. $\mathrm{Cov}(X,Y)$ is the covariance of them in practice. Covariance and standard deviation can only be estimated from a limited number of samples. Therefore, the sample of PCC is

$$r_{X,Y} = \frac{\sum_{i=1}^{n}(X_i - \overline{X})(Y_i - \overline{Y})}{\sqrt{\sum_{i=1}^{n}(X_i - \overline{X})^2 (Y_i - \overline{Y})^2}} = \frac{1}{n-1}\sum_{i=1}^{n}\left(\frac{X_i - \overline{X}}{s_X}\right)\left(\frac{Y_i - \overline{Y}}{s_Y}\right) \tag{2}$$

$$\overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i \tag{3}$$

$$s_X = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})^2} \tag{4}$$

where $r_{x,y}$ is the correlation coefficient, and $\bar{X}$ and $\bar{Y}$ are the mean values of the variables X and Y , respectively. n is the number of samples. The value of $r_{x,y}$ is among $[-1,1]$. The closer the absolute value of $r_{x,y}$ to 1 is, the higher the correlation between the variables X and Y .

2) Nonlinear Correlation Analysis Based on ANNCA: An artificial neural network (ANN) is an algorithmic mathematical model that imitates the characteristics of an animal neural network and carries out distributed parallel information processing. An ANN is formed by stacking single-layer or multilayer neurons. Each layer is connected by weight and bias, and the activation function makes the ANN have the ability of nonlinear representation. The goal of an ANN is to establish the mapping model between the input and the output, and to minimize the distance between the model output and the real output. The flow direction of data in the ANN contains forward propagation and backpropagation. Forward propagation is used to pass the information of the input to each neuron in the network. Backpropagation is to reversely update the parameters in the model according to the loss between the model output and the real output. To establish the neural network model more quickly and stably, the stochastic gradient descent method is often used for backpropagation.

ANNCA can be divided into two parts: ANN and PCC. If there is a correlation among two variables X and Y , the correlation can be expressed by $Y = f(X)$ or $X = f(Y)$.

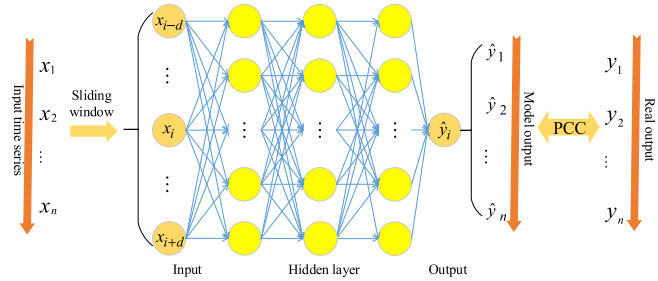

Fig. 1. ANNCA structure.

Based on this idea, X and Y are taken as the input and output of the neural network, respectively. A large number of X and Y samples are used to train the neural network model. If the input X without participating in the training can accurately predict the corresponding Y , it is considered that there is a prominent correlation between X and Y . The PCC between the model output and the real output is used to measure this relationship. The ANNCA framework is shown in Fig. 1.

In Fig. 1, x_i is the value of variable *X* at time *i*. y_i is the value of variable *Y* at time *i*. $\hat{y}_i$ is the predicted value of variable *Y* at time *i*. *d* is the delay length. The implementation of ANNCA is summarized as follows. First, the input variable X and output variable Y are, respectively, used to obtain the sample dataset by sliding window operation. In particular, the $i\mathrm{th}$ sample has an input of $[x_{i-d}, x_{i+d}]$ and an output of y_i . The data of $[x_{i-d}, x_{i-1}]$, x_i , and $[x_{i+1}, x_{i+d}]$, respectively, represent the past, present, and future data of variable X at time i . Second, the dataset is divided into training, validation, and testing sets. The training and validation sets are used to train and adjust the ANN model of ANNCA. The test set is used to generate $\hat{y}$ from the ANN model. Note that the samples in the test dataset need to be time-continuous data. Finally, the PCC value between Y and $\hat{Y}$ is calculated as the degree of correlation between variables X and Y . The result of ANNCA is among $[-1,1]$. The closer the absolute value of result to 1 is, the higher the correlation between the variables X and Y is.

ANNCA comprehensively considers the past, present, and future information of two variables, which has powerful nonlinear processing capabilities and can effectively extract spatio-temporal correlation information. Different from PCC, ANNCA analysis results may be related to the positions of X and Y . ANNCA analysis results of X and Y may be different from those of Y and X .

B. Long Short-Term Memory (LSTM)

LSTM was first proposed by Hochreiter and Schmidhuber [29], which can be regarded as a special recurrent neural network (RNN). Theoretically, an RNN should learn all the dependencies from the historical data. However, with the increase of network layers, an RNN has problems, such as gradient explosion and gradient disappearance, so LSTM comes into being. Compared with an RNN, the structure of LSTM is more complex. The LSTM model often contains an input layer, an output layer, and some hidden layers. There are many layers in the hidden layers,

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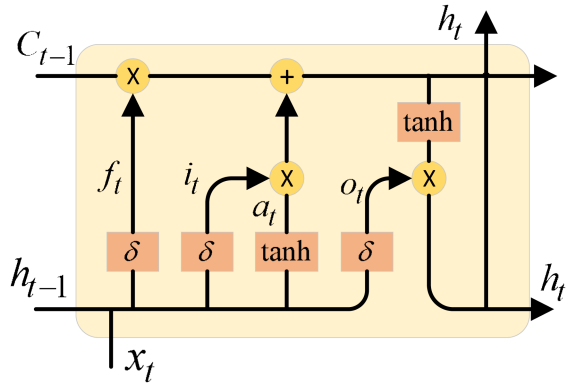

Fig. 2. LSTM cell structure.

and each layer is composed of three gates and a storage state, which are forget gate, input gate, output gate, and memory cell, respectively. The structure of LSTM is shown in Fig. 2.

The forget gate is the first to receive the information and controls the amount of historical information flowing into the current state. Its expression is as follows:

$$f_t = \delta\left(W_f h_{t-1} + U_f x_t + b_f\right). \tag{5}$$

The input gate is used to update the current input data and determines which data are entered into the memory cell. It can be expressed as follows:

$$i_t = \delta\left(W_i h_{t-1} + U_i x_t + b_i\right) \tag{6}$$

$$C_t = f_t \cdot C_{t-1} + i_t \cdot a_t. \tag{7}$$

The output gate controls the output of the model and the information flowing into the next hidden layer in the memory unit of the current LSTM. It can be expressed as follows:

$$o_t = W_o h_{t-1} + U_o x_t + b_o \tag{8}$$

$$h_t = o_t \cdot \tanh\left(C_t\right). \tag{9}$$

The memory cell determines the information currently stored and flowing to the next layer. It can be expressed as follows:

$$a_t = \tanh\left(W_a h_{t-1} + U_a x_t + b_a\right). \tag{10}$$

In the above formulas, W_f and U_f represent the weights of forget gate. W_i and U_i represent the weights of the input gate. W_o and U_o represent the weights of output gate. W_a and U_a represent the weights of the memory cell. b_f , b_i , b_o , and b_a represent the bias of forget gate, input gate, output gate, and memory cell, respectively. h_{t-1} represent the value of the hidden state of the last layer. δ represent the sigmod function. x_t is the input information of the current moment.

C. Framework of Anomaly Detection Based on STC-LSTM

This article proposes an STC-LSTM framework for the detection and recovery value prediction of UAV flight data. The process is shown in Fig. 3.

The specific process of STC-LSTM is as follows. First, flight data are collected from the UAV. Second, the data are preprocessed. Third, ANNCA and PCC are used to mine the correlation in flight data, and to establish correlation parameter sets. The input and the output parameters of the LSTM model are selected according to the correlation parameters sets. Fourth, the flight data are reconstructed to make the dataset. The dataset is divided into training, validation, and testing sets. Training and validation sets are used to train and adjust the LSTM model. The test set containing anomaly data is used to obtain the defined range of anomaly and to verify the performance of the STC-LSTM. Finally, the anomaly detection and recovery prediction of UAV flight data are realized.

III. Experiment Results and Analysis

A. Data Description

1) Flight Data Collection: The experimental data used in this article are all collected from the actual flight data of UAV, which is shown in Fig. 4.

The UAV is a "Clouds" fixed-wing airplane with a wingspan of 1880 mm, fuselage length of 960 mm, fuselage thickness of 260 mm, weight of 850 g, payload of 600–1200 g, and cruising speed of 50–80 km/h. The UAV flight data are collected by Pixhawk 4. The collected data include 63 sensor data, such as pitch angle, roll angle, pitch angle rate, roll angle rate, yaw angle rate, sky speed, airspeed, ground speed, and altitude. A total of 37 flight data are collected. The maximum flight time is 15 min. The minimum is 10 min. The flight tasks include take off, level flight, turn, head up, landing, and other operations.

2) Data Preprocessing: The data obtained from UAV have some problems, such as missing value, repeated value, and inconsistent sampling frequency. The flight data need to be preprocessed. Aiming at the problem of data missing, the linear interpolation is used to fill in the missing data. The method of deleting repeated timestamps is used to solve the problem of repeated records. Due to the different sensors to collect the UAV flight data, the sampling frequency is inevitably different. This article uses the resampling method to unify the UAV flight data on the timeline, and the sampling interval of the final sample is 30 ms. In addition, the data need to be normalized. The data processed through the above steps can be used for experimental research.

3) Anomaly Injection: Under real conditions, the anomaly in the operation of the system is rare and difficult to collect. Therefore, in the sample set, the number of normal samples is far greater than the number of anomaly samples. However, anomaly samples are needed to verify the STC-LSTM method. In this article, anomaly samples are generated by injecting anomalies in normal data. Three kinds of anomaly injection methods are used in this article, which are deviation anomaly, static anomaly, and drift anomaly, respectively [23], [24].

Deviation anomaly: $y_f(t) = y(t) + \beta$, where β is a constant. $y(t)$ is normal data and t is a period of time.

Static anomaly: $y_f(t) = \beta$, where β is a constant. t is a time period. The static anomaly shows that the flight data are a constant value in a time period.

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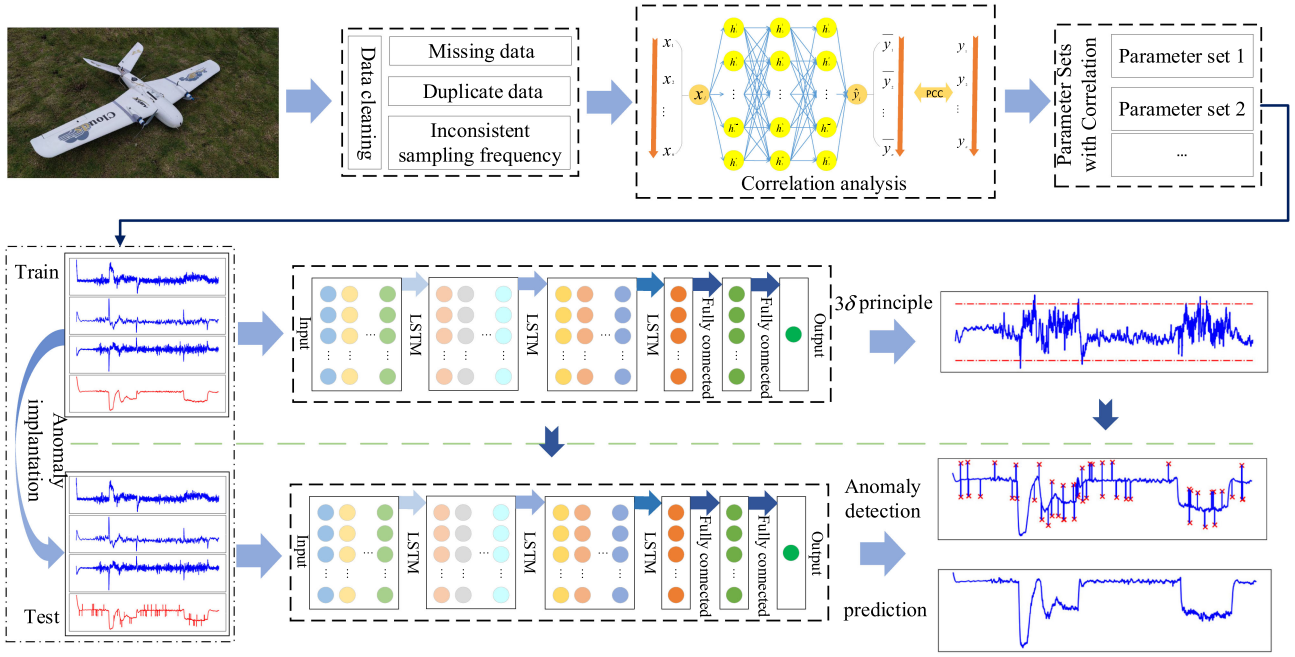

Fig. 3. Framework of anomaly detection based on STC-LSTM.

Fig. 4. "Clouds" fixed-wing UAV.

TABLE I
FLIGHT PARAMETERS

S/N	Flight parameters	S/N	Flight parameters
0	Pitch rate	9	Ground speed
1	Roll rate	10	Axial overload
2	Yaw rate	11	Pitch angle
3	Normal overload	12	Roll angle
4	North speed	13	Sideslip Angle
5	East speed	14	Right aileron command
6	Sky speed	15	Left aileron command
7	Calibrated airspeed	16	Altitude
8	True airspeed	17	Barometric altitude

Point anomaly: $y_f(t) = y(t) + \beta$. Different from the deviation anomaly, the point anomaly occurs at multiple independent time points, whereas the deviation anomaly occurs within a time period.

Roll angle is one of the important parameters to reflect the safe operation of UAVs. This article takes roll angle as a detected parameter. A section of flight data with a length of 25 000 is selected as the analysis object. The normal data of roll angle are shown in Fig. 5(a). The anomaly dataset of roll angle is generated manually by the above anomaly injection methods. Fig. 5(b) shows the deviation anomaly, where the anomaly range is $[7000, 20000]$ and anomaly degree of deviation anomaly is the difference between the maximum and minimum of normal data timing 0.2. Fig. 5(c) shows the static anomaly, where the anomaly data are stuck at $-7°$ and the range is $[10200, 18500]$. Fig. 5(d) shows the point anomalies, which contains 50 random point anomaly and anomaly degree is the difference between the maximum and minimum of normal data timing 0.2.

B. Correlation Analysis of High-Dimensional Flight Data

UAV flight data contain 63 parameters, and some parameters adopt the redundant design. In the analysis of correlation, redundant parameters are removed. A total of 18 parameters (shown in Table I) need to be analyzed for correlation. Based on data preprocessing, PCC and ANNCA are, respectively, used for correlation analysis of flight data. The analysis results are shown in Figs. 6 and 7.

PCC can only find the linear correlation in data. ANNCA can find both linear and nonlinear correlations. The result of PCC and ANNCA are among $[-1, 1]$. For the convenience of comparison, the values of ANNCA and PCC results are taken absolutely, respectively, so that the result is among $[0, 1]$.

Figs. 6 and 7, respectively, show the correlation analysis results between PCC and ANNCA. The flight parameters corresponding to the serial number of coordinate axes are shown in Table I. The values in Figs. 6 and 7 is the correlation among two parameters, and the value is among $[0, 1]$. Note that a high value

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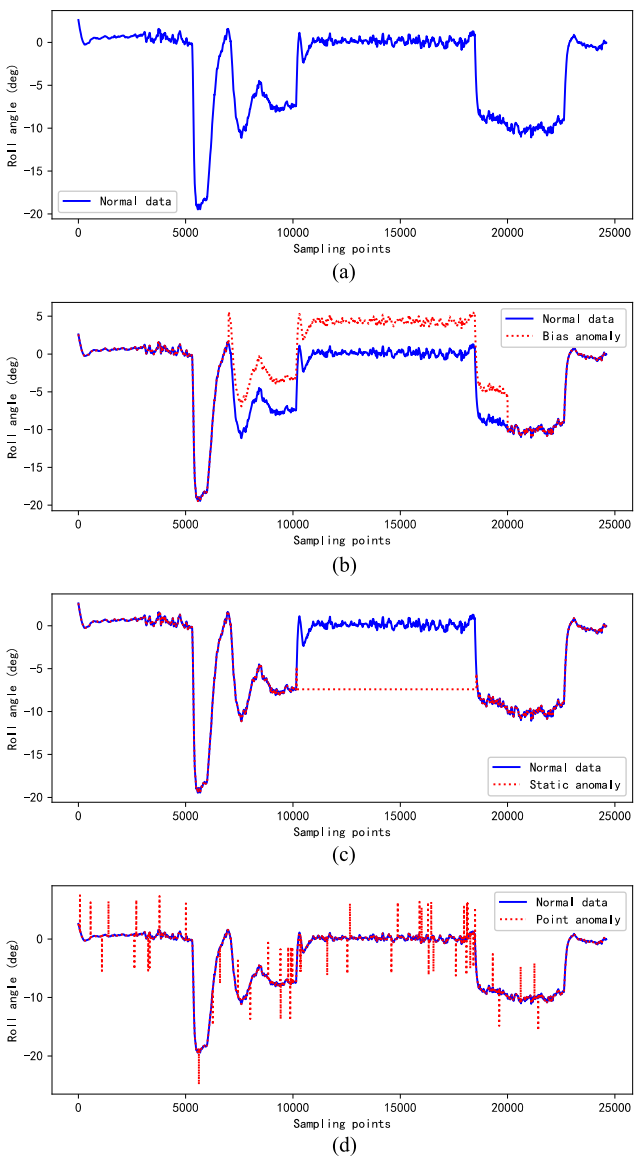

Fig. 5. Anomaly injection. (a) normal data. (b) Implantation of bias anomaly. (c) Implantation of static anomaly. (d) Implantation of point anomaly.

indicates a high correlation of two parameters. The results of PCC analysis were symmetric with the main diagonal, whereas ANNCA analysis was asymmetric. The reason is that the result of ANNCA was related to the positional order of parameters, whereas PCC does not need to consider the positional order of parameters.

Generally, if the value of PCC is greater than 0.6, it is considered that the correlation among two parameters is strong. This article uses 0.6 as the threshold for the strong correlation for PCC and ANNCA. Comparing Figs. 6 with 7, the flight parameter pairs that have a strong correlation in the PCC results also have a strong correlation in ANNCA. However, ANNCA contains some correlations that do not exist in PCC, and it indicates that ANNCA can find more abundant correlations than PCC.

In this article, 0.6 is used as the threshold for parameter selection of ANNCA and PCC to establish the correlation parameter

TABLE II
CORRELATION PARAMETER SET OF THE ROLL ANGLE

Correlation parameters	PCC	ANNCA
Pitch angle rate	0.34	**0.87**
Roll rotation rate	0.06	**0.96**
Yaw angle rate	**0.98**	**0.99**
Normal overload	0.26	**0.67**
Sideslip angle	**0.65**	**0.94**
Right aileron command	0.29	**0.98**
Left aileron command	0.29	**0.97**

sets. Taking the roll angle as an example, in the analysis results of ANNCA, there are seven parameters (shown in Table II) that have a correlation with the roll angle greater than 0.6. But in the result of PCC, there are only two parameters (shown in Table II) that have a correlation with the roll angle greater than 0.6. In this article, the results of ANNCA are used to establish an anomaly detection and recovery value prediction model. Such condition means there are seven parameters in the correlation parameter set of roll angle, which are pitch angle rate, roll rotation rate, yaw angle rate, normal overload, sideslip angle, right aileron command, and left aileron command.

The method of parameter selection using correlation analysis can effectively reduce the dependence on expert knowledge. Taking UAV anomaly detection as an example, even if researchers do not understand the specific meaning of each flight parameter and UAV system structure, they can effectively carry out anomaly detection research on UAV flight data. The knowledge of a specific field is abstracted through correlation analysis and transforms into a general data analysis problem.

C. Anomaly Detection and Recovery Prediction Based on STC-LSTM

1) Model Training With Spatio-Temporal Correlation: The core of STC-LSTM is to establish the anomaly detection and recovery prediction model by mining the spatio-temporal correlation information of correlation parameter sets. In particular, STC-LSTM is used to predict the current data of the detected parameter through the historical data of correlation parameter sets. Unlike STC-LSTM, ANNCA makes comprehensive use of historical, present, and future flight data. However, in online anomaly detection, present and future flight data cannot be used for analysis, and only historical data can be used for analysis for STC-LSTM.

Before training the model, the data need to be reconstructed so that it can meet the input and output of the model. The formula is expressed as follows:

$$X(t) = \begin{bmatrix} x_1(t-1), x_1(t-2) \cdots x_1(t-D) \\ x_2(t-1), x_2(t-2) \cdots x_2(t-D) \\ \vdots \\ x_n(t-1), x_n(t-2) \cdots x_n(t-D) \end{bmatrix} \tag{11}$$

$$Y(t) = y(t) \tag{12}$$

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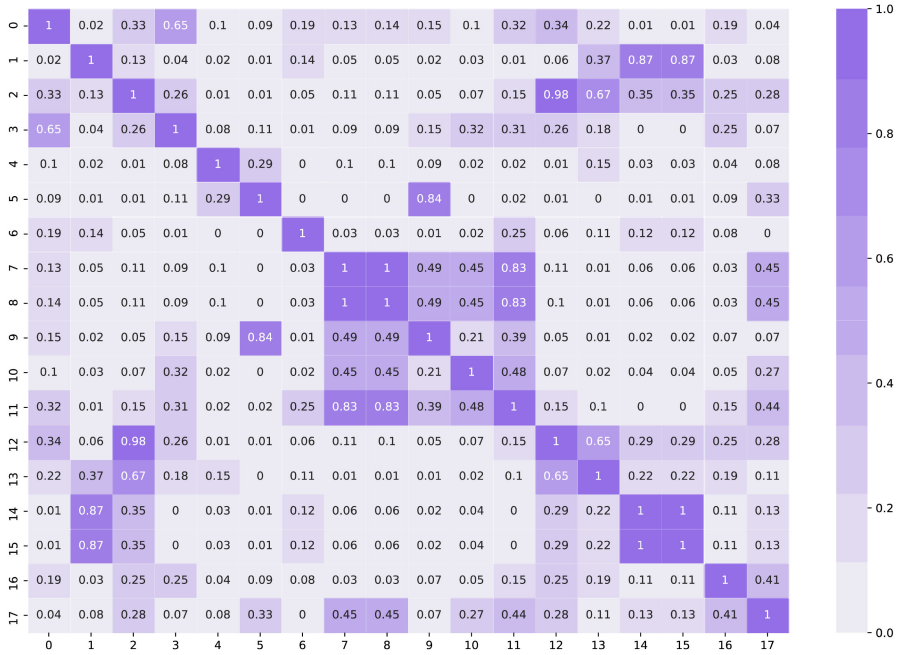

Fig. 6. Parameter sets of PCC.

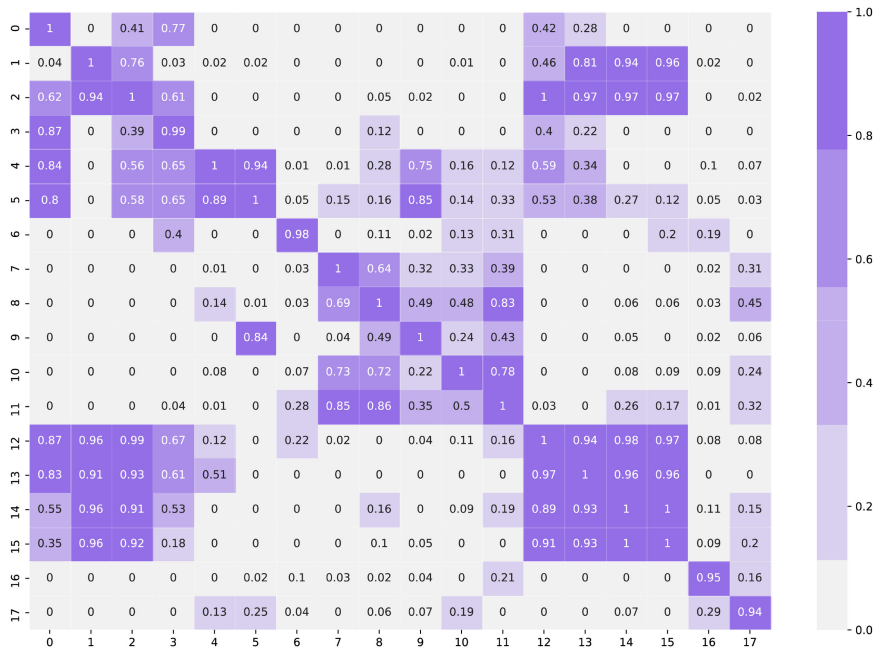

Fig. 7. Parameter sets of ANNCA.

where $X(t)$ is the input of the model, and its dimension is $n \times D$. n is the number of parameters in the correlation parameter set. D is the length of the sliding window. $x_n(t-1)$ is the data of the correlation parameter at time t . $Y(t)$ is the target value of the model output. $y(t)$ is the data of the detected parameter at time t . The actual collected flight data are made into a training set and a test set in the form of $X(t)$ and $Y(t)$. In the process of training the model, the training set $X(t)$ is the input of the model, and the output $\hat{Y}(t)$ is obtained, which is the predicted value of the detected parameter. The model is trained to minimize the difference between $\hat{Y}(t)$ and $Y(t)$. This article mainly builds the anomaly detection and recovery prediction model of STC-LSTM based on LSTM. The model contains three LSTM layers and two fully connected layers. The input dimension of the model is $D \times n$, and the output dimension is 1. The LSTM model is shown in Fig. 8.

2) Anomaly Discrimination: This article uses the 3σ principle to locate the anomaly points [30]. First, we input $X(t)$ (data

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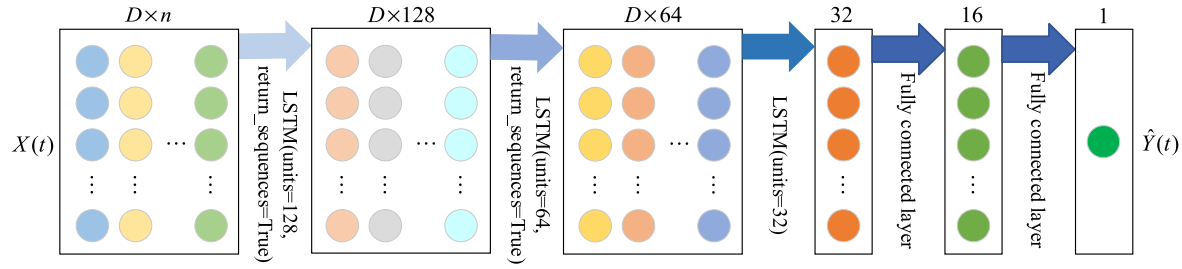

Fig. 8. LSTM model.

before the anomaly are implanted) in the test dataset into the trained model to get $\hat{Y}(t)$. Second, the residual $R(t)$ between $\hat{Y}(t)$ and $Y(t)$ is calculated as follows:

$$R(t) = \hat{Y}(t) - Y(t). \tag{13}$$

Third, the mean M and standard deviation S are calculated as follows:

$$M = \frac{1}{n}\sum_{t=1}^{n} R(t) \tag{14}$$

$$S = \sqrt{\frac{1}{n-1}\sum_{t=1}^{n}(R(t) - M)^2}. \tag{15}$$

Finally, the allowable upper and lower limits of $R(t)$ are calculated as follows:

$$C = M + 3 \times S \tag{16}$$

$$F = M - 3 \times S \tag{17}$$

where C is the upper limitation of 3σ principle. F is the lower limitation of the 3σ principle. If $R(t)$ is within the 3σ interval, $Y(t)$ is considered normal data, otherwise, $Y(t)$ is considered anomaly data.

3) Anomaly Detection Evaluation Criteria and Results: In anomaly detection, the main concern is the ability of the model to find anomalies. In this article, false positive rate (FPR), false negative rate (FNR), and accuracy (ACC) are used as evaluation metrics, which are given as follows:

$$\mathrm{FPR} = \frac{\mathrm{FP}}{\mathrm{FP} + \mathrm{TN}} \tag{18}$$

$$\mathrm{FNR} = \frac{\mathrm{FN}}{\mathrm{TP} + \mathrm{FN}} \tag{19}$$

$$\mathrm{ACC} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{FP} + \mathrm{TP} + \mathrm{TN} + \mathrm{FN}} \tag{20}$$

where TN is true negative, FN is false negative, FP is false positive, and TP is true positive. FPR is the proportion of all anomaly data detected as normal by the model, reflecting the probability that the model misses reporting when anomalies occur in the system. Note that a small FPR indicates good performance of anomaly detection and is desired. FNR is the proportion of all normal data detected as anomalies by the model, reflecting the probability of false positives of the model when the system is in normal operation. Note that a small FNR indicates good performance of anomaly detection and is desired. ACC is a comprehensive index that reflects the ability of the model to make correct actions. Note that a large ACC indicates good performance of anomaly detection and is desired.

Fig. 9 shows the results of anomaly detection using the STC-LSTM method. The yellow line is the output of the model, which represents the prediction value of the real output. The blue line is the real output value of the roll angle. The pink band is the upper and lower limit of the 3δ principle, which is distributed on both sides of the yellow line. If the blue line is not inside the pink band, the data are considered to be abnormal.

Fig. 9(a) shows the results of anomaly detection on normal data. FPR is 0 in the result, indicating that STC-LSTM does not detect normal data as anomaly. Fig. 9(b) shows the results of the detection of the deviation anomalies. STC-LSTM detected all 13 000 deviation anomalies. Fig. 9(c) shows the results of the detection of static anomalies detection. STC-LSTM detected all 8300 deviation anomalies. Fig. 9(d) shows the results of the detection of point anomalies. STC-LSTM detected all 50 deviation anomalies.

It can be seen intuitively from Fig. 9 that STC-LSTM can accurately find and locate data anomalies. To verify the superiority and effectiveness of STC-LSTM, k nearest neighbor (KNN) [22], support vector regression (SVR), and LSTM are selected for comparison. KNN is the temporal correlation algorithm, which builds an anomaly detection model by using the historical data of the detected parameters. The input size of KNN is 400 and the number of neighbors is 20. There are few anomaly detection methods only using spatial correlation in flight data, and the input parameters of SVR are selected with the results from ANNCA. The input size of SVR is n , where n is the number of parameters in the correlation parameter set. The output size is 1. In an LSTM method, 63 parameters are directly used to train the LSTM model without parameter selection. In the following, we will evaluate the four methods of STC-LSTM, LSTM, SVR, and KNN in combination with the three metrics of FNR, FPR, and ACC.

Table III shows the anomaly detection results of the four algorithms. "/" represents a null value. The KNN algorithm is a temporal correlation anomaly detection method. For instantaneous anomalies, such as point anomalies, the duration of the anomaly is very short, and it has a good detection effect. FNR, FPR, and ACC are all maintained at a good level. But for continuous anomalies, such as static anomalies and deviation anomalies, where the anomaly lasts for a long time, KNN cannot

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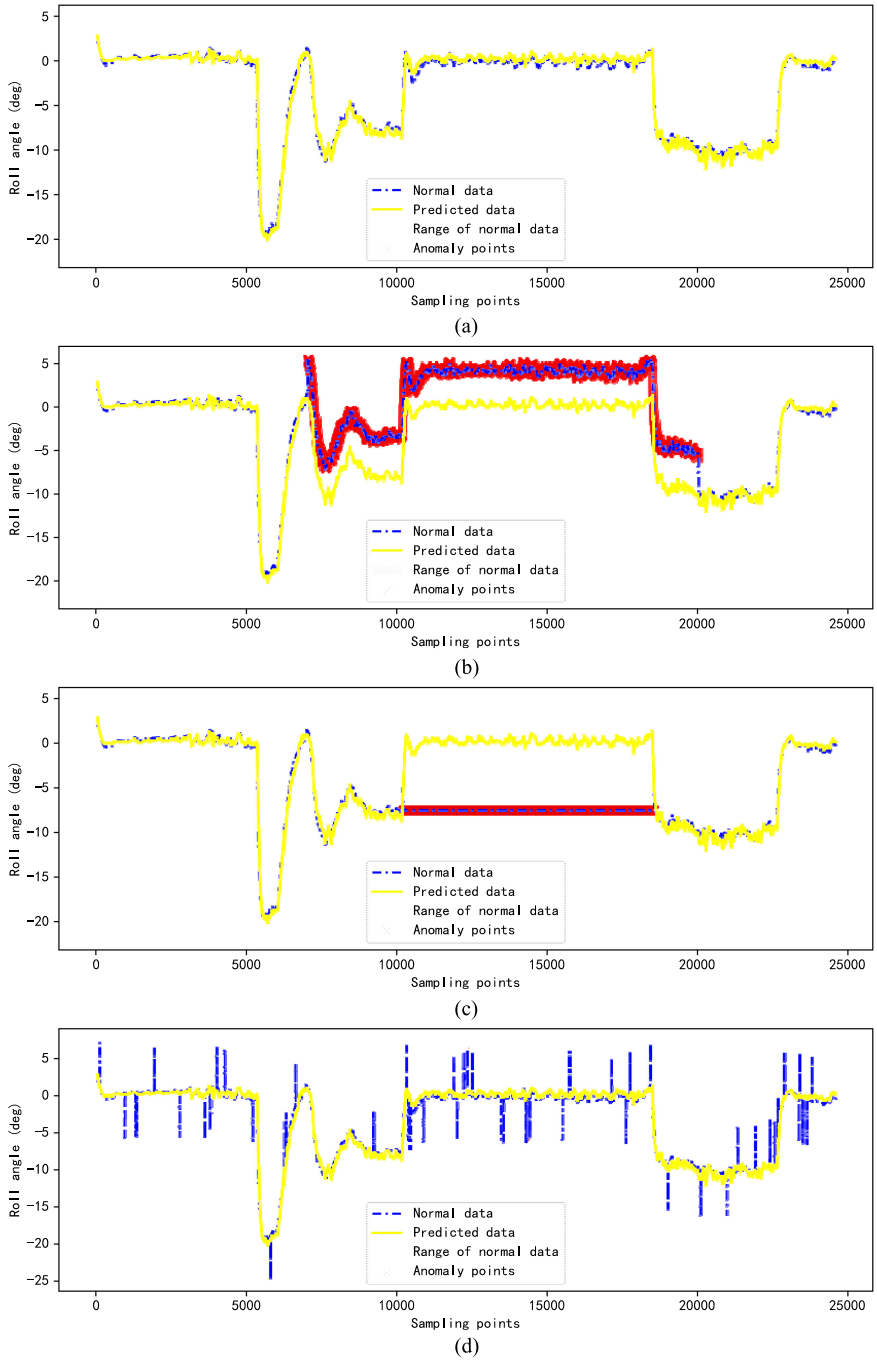

Fig. 9. Anomaly detection results of three types of anomaly data in our method. (a) normal data. (b) Deviation anomaly. (c) Static anomaly. (d) Point anomaly.

find anomalies in the data. The SVR algorithm is a spatial correlation algorithm, and its core idea is multiple regression. For deviation anomalies, static anomalies, and point anomalies, FPR is 2.5%, 2.0%, and 0, respectively, which indicates that SVR has good anomaly detection capabilities. However, for detecting data without anomalies implanted, the SVR algorithm produces 17.3% of false detections. The LSTM method does not execute parameter selection, and all 63 parameters are used to train the model. For deviation anomalies, static anomalies, and point anomalies, FNR is 0 and FPR is 1, which means that LSTM judges all data as normal data and cannot find anomaly data. From the results of LSTM model training, the LSTM model is underfitting due to too many parameters, and the output of the model is always 0, which proves that the LSTM method has no ability of anomaly detection. Compared with KNN, SVR, and LSTM, STC-LSTM greatly improves accuracy in anomaly detection. For no anomalies, point anomalies, static anomalies, and deviation anomalies, the FNR and FPR are 0 and ACC is 1, which means that STC-LSTM can perfectly find anomalies in the flight data and avoid causing missed detection. Compared with LSTM, parameters in flight data of STC-LSTM are reduced by 88.7%. STC-LSTM defines the internal relationship of flight parameters

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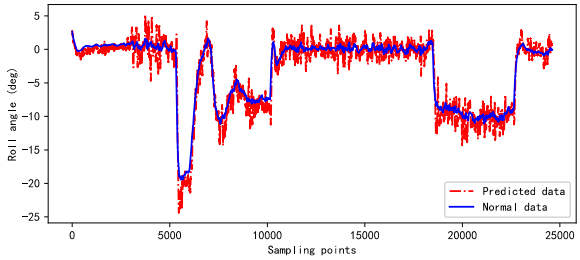
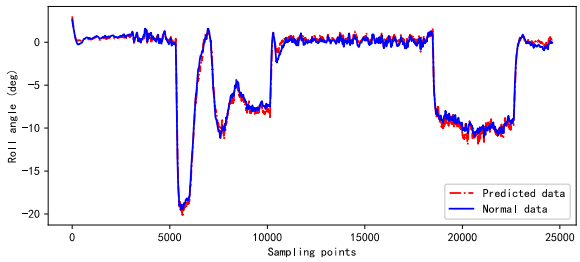


(a)



(b)

Fig. 10. Comparison of prediction effects between SVR and STC-LSTM. (a) SVR recovery value prediction. (b) STC-LSTM recovery value prediction.

TABLE III
COMPARATIVE EVALUATION OF THREE METHODS: KNN, SVR, AND STC-LSTM

Model	Anomaly	*FNR*	*FPR*	*ACC*
KNN	No	0.5%	/	99.5%
	Deviation	1.2%	96.0%	48.7%
	Static	0.8%	100.0%	65.5%
	Point	0.5%	**0**	99.5%
SVR	No	17.3	/	82.7%
	Deviation	18.2%	2.5%	90.1%
	Static	19.7%	2.0%	86.9%
	Point	17.3%	**0**	82.8%
LSTM	No	**0**	/	**100.0%**
	Deviation	**0**	100.0%	47.2%
	Static	**0**	100.0%	98.0%
	Point	**0**	100.0%	99.7%
STC-LSTM	No	**0**	/	**100.0%**
	Deviation	**0**	**0**	**100.0%**
	Static	**0**	**0**	**100.0%**
	Point	**0**	**0**	**100.0%**

and provides certain interpretability. Reducing parameters in flight data can greatly reduce the amount of calculation, and reduce the difficulty of troubleshooting and maintenance.

4) Recovery Value Prediction: The anomaly detection of UAV flight data can minimize the impact of anomalies. In addition to timely detection of anomalies and accurate positioning of anomalies when anomalies occur, it is also necessary to predict the recovery value of anomalies. The recovery value prediction can help the operator better grasp the flight status of the UAV and execute emergency plans when an anomaly occurs. Meanwhile, for closed-loop control of the UAV, the recovery prediction can convert anomaly data into normal data in time. In this article, we use mean square error (MSE) and mean absolute error (MAE) to evaluate the ability of STC-LSTM to predict the recovery value as follows:

$$\mathrm{MSE} = \frac{1}{n}\sum_{i=1}^{n}(\hat{y}_i - y_i)^2 \tag{21}$$

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}|\hat{y}_i - y_i| \tag{22}$$

where $\hat{y}_i$ is the output of the model at time i , and y_i are real data. There are two valuation metrics that reflect the ability of the prediction data to track the actual data. Consider that the values of the model are normalized. The closer the value of the three evaluations index is to 0, the better the accuracy of the model will be.

When the detected parameter is an anomaly or completely invalid, the recovery value of the anomaly data can be predicted through correlation parameters. Among the above three anomaly detection methods, SVR, LSTM, and STC-LSTM can be exploited for spatial correlation and used to recovery prediction, but the LSTM method is underfitting. This article chooses SVR and STC-LSTM to test the recovery value prediction.

Fig. 10 shows the visualization of recovery value prediction. On the whole, both SVR and STC-LSTM algorithms can follow the changing trend of real data. However, from a local point of view, there is much noise in the prediction data of SVR, which cannot accurately match the real data. In contrast to the STC-LSTM, the degree of overlap between the predicted data and the real data is big, the noise in the predicted data is small, and accurate prediction can be achieved.

It can be seen from Table IV that in terms of recovery value prediction, the three metrics of STC-LSTM are significantly lower than those of SVR. MSE and MAE of STC-LSTM are 0.000343 and 0.014651, respectively. Compared with SVR, the MSE and the MAE of STC-LSTM are reduced by 92.1% and 70.0%, respectively. It shows that STC-LSTM can accurately predict the recovery value of anomalies.

TABLE IV
SVR AND STC-LSTM EVALUATION RESULTS

Model	*MSE*	*MAE*
SVR	0.004346	0.048799
STC-LSTM	0.000343	0.014651

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IV. Conclusion

Aiming at the problem of anomaly detection and recovery value prediction of UAV flight data, this article designed an STC-LSTM framework.

First, ANNCA was used to find the correlation in the data and obtain the correlation parameter set, which was used as the basis for the selection of the input parameters of the STC-LSTM model. Second, the STC-LSTM model was used to mine the spatio-temporal correlations among multiple flight parameters, and the 3σ principle was used as the basis for anomaly determination. Then, three types of abnormal samples were made by the method of anomaly injection. Finally, anomaly sample sets were used to test the STC-LSTM method. Meanwhile, STC-LSTM could also predict the recovery value of abnormal data.

In this article, the flight data collected from a fixed-wing UAV were used for experiments. The detection results of STC-LSTM were FPR $= 0$, FNR $= 0$, and ACC $= 1$ for deviation, static, and point anomaly. For normal data, there was no false detection. For anomaly data recovery value prediction, MSE and MAE were 0.000343 and 0.014651, respectively, which were 92.1% and 70.0% lower than SVR method. Compared with LSTM, parameters of STC-LSTM were reduced by 88.7%. Through the above-mentioned results of experiments, it was proved that using the correlation method to select parameters could effectively avoid the dependence on expert knowledge and greatly reduce the parameters of model. Meanwhile, STC-LSTM could accurately locate the anomalies of UAV flight data and provide high-precision recovery prediction values.

In the future, we will consider applying this method to an embedded platform and organically combine it with flight control of UAV to realize real-time anomaly detection.

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Jie Zhong received the B.Eng. degree in mechanical engineering from the College of Mechanical Engineering, Chengdu University of Technology, Sichuan, China, in 2020. He is currently working toward the M.S. degree in aerospace science and technology with the School of Aeronautics and Astronautics, Sichuan University, Chengdu, China.

His research interests include anomaly detection and deep learning.

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Yujie Zhang (Member, IEEE) received the B.S. degree in instrumentation science and technology from the Harbin Institute of Technology at Weihai, Weihai, China, in 2014, and the Ph.D. degree in information and communication engineering from the Harbin Institute of Technology, Harbin, China, in 2021.

He is currently an Assistant Professor with the College of Electrical Engineering, Sichuan University, Chengdu, China. His current research interests include prognostics and health management, condition monitoring, data-driven degradation modeling, and electronic–mechanical system simulation.

Jianyu Wang received the B.Eng. degree in mechanical engineering from the Chongqing Technology and Business University, Chongqing, China, in 2017, and the M.S. degree in aeronautical and astronautical science and technology in 2020 from Sichuan University, Chengdu, China, where he is currently working toward the Ph.D. degree in mechanical engineering with the School of Aeronautics and Astronautics.

His research interests include mechanical fault diagnosis and deep learning.

Chong Luo received the B.Eng. degree in mechanical engineering from the Chongqing Technology and Business University, Chongqing, China, in 2017, and the M.S. degree in aeronautical and astronautical science and technology from Sichuan University, Chengdu, China, in 2021. He is currently working toward the Ph.D. degree with the School of Electrical Engineering, Sichuan University, Sichuan, China.

His current research interests include cyclostationary analysis, mechanical fault diagnosis, and unmanned aerial vehicle anomaly detection.

Qiang Miao (Senior Member, IEEE) received the B.E. degree in reliability engineering and the M.S. degree in mechanical engineering from the Beijing University of Aeronautics and Astronautics, Beijing, China, in 1998 and 2001, respectively, and the Ph.D. degree in industrial engineering from the University of Toronto, Toronto, ON, Canada, in 2005.

He is currently a Professor in prognostics and health management with the College of Electrical Engineering, Sichuan University, Chengdu, China. In 2011, he was selected as the New Century Excellent Talents in University and the Reserve Candidate of Sichuan Province Academic and Technology Leadership. In 2012, he was appointed as the Adjunct Professor with the City University of Hong Kong. In 2013, he was selected as the Excellent Expert of Sichuan Province with Great Accomplishments. In the past years, he received more than 20 research grants, and authored or coauthored more than 80 research papers and holds nine patents. His current research focuses on reliability and machinery condition monitoring and health assessment.

Dr. Miao has been a Senior Member of the IEEE Reliability Society since 2012. He was the recipient of the Ministry of Education Natural Science Award (Second Class) in 2009.

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